Deep Learning

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A few points before starting

- **Mark Distribution**
  1. Midsem - 30
  2. Endsem - 50
  3. One Class test - 10 (towards March end)
  4. Assignment (1) and ACL (1) - 10
- Attendance as per institute norms
- Relative grading will be followed
- 1 TA is assigned for each 20 students
- Google classroom will be created for sharing the materials and assignments.
- Prerequisites: Basic linear algebra, Concept of gradients, Probability (only where required). **You don't need to be a maths expert.**
- **IMP** - Brush-up your concepts on Machine Learning

Learning Outcomes

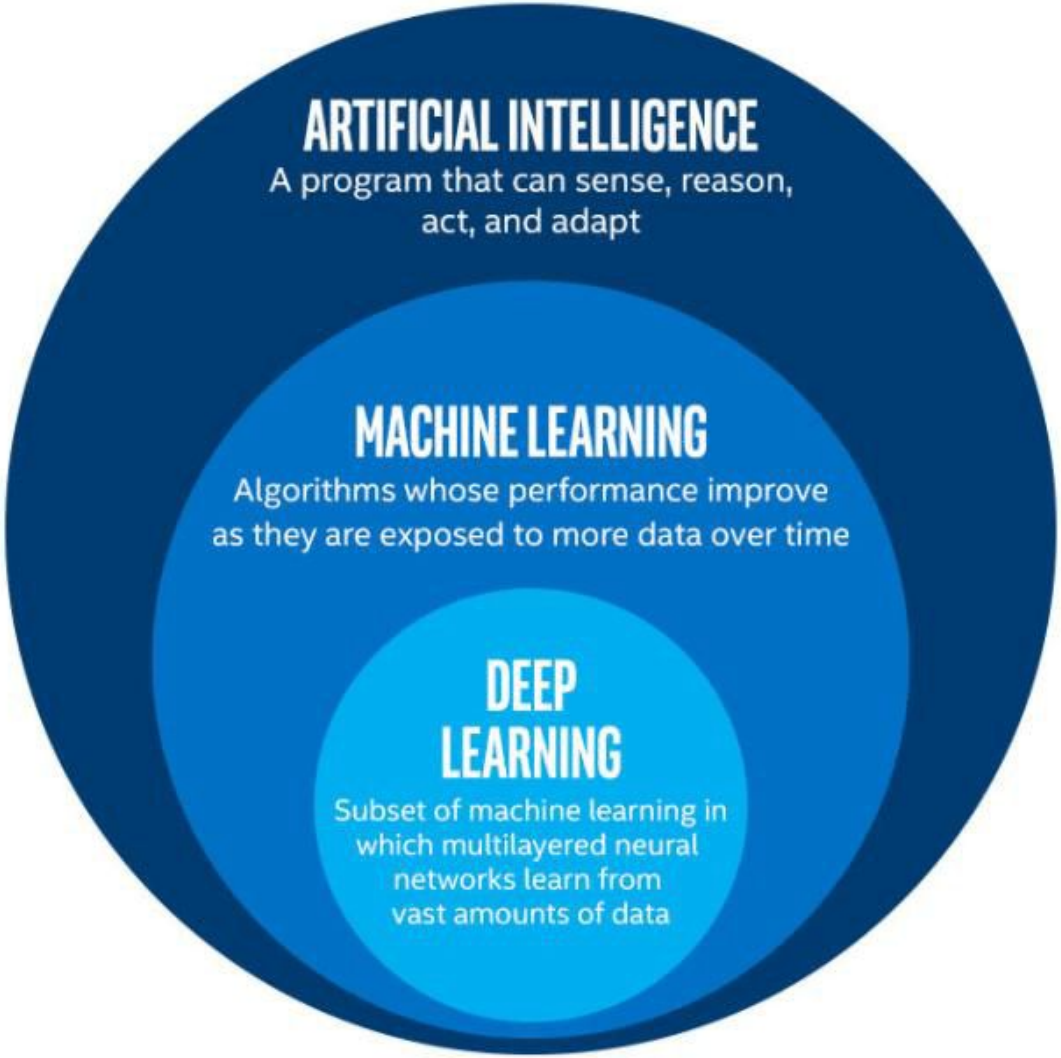
- Describe the basic principles and mathematical foundations of deep learning models, including neural networks, loss functions, backpropagation, etc.
- Apply key theoretical concepts such as Gradient Descent optimization techniques, activation functions, and learning algorithms to understand how deep models learn.
- Derive and interpret the mathematical formulations behind key neural network algorithms such as Multi Layer Perceptron (MLP), Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Generative models, Transformers, etc.
- Analyse the internal behaviour of deep networks with respect to convergence, expressiveness, overfitting, and generalization using mathematical and conceptual tools.
- Evaluate different state-of-the-art deep learning architectures based on their theoretical properties, limitations, and use cases.

Books

- **Text Books**
  - Ian J. Goodfellow, Yoshua Bengio, and Aaron Courville, Deep Learning, MIT Press, 1st Edition (2016), ISBN: 978-0262035613
  - Simon J.D. Prince, Understanding Deep Learning, MIT Press, 1st Edition (2023), ISBN: 978-0262048644
  - Charu C. Aggarwal, Neural Networks and Deep Learning: A Textbook, Springer, 2nd Edition (2023), ISBN: 978-3-031-29641-3

Books

- **Reference Books**
  - Kevin P. Murphy, Machine Learning: A Probabilistic Perspective, MIT Press, 1st Edition (2012), ISBN: 978-0262018029
  - Trevor Hastie, Robert Tibshirani, and Jerome Friedman, The Elements of Statistical Learning: Data Mining, Inference, and Prediction, Springer, 2nd Edition (2009), ISBN: 978-0387848570
  - Aston Zhang, Zachary C. Lipton, Mu Li, and Alexander J. Smola, Dive into Deep Learning, Self-published (Open Source Textbook), Version 1.0, Apache License 2.0, URL: https://d2l.ai



AI: Making machines behave intelligently

ML: Learning patterns from historical data

DL: Learning *hierarchical representations* using neural networks

Deep Learning

— It is inspired by the brain

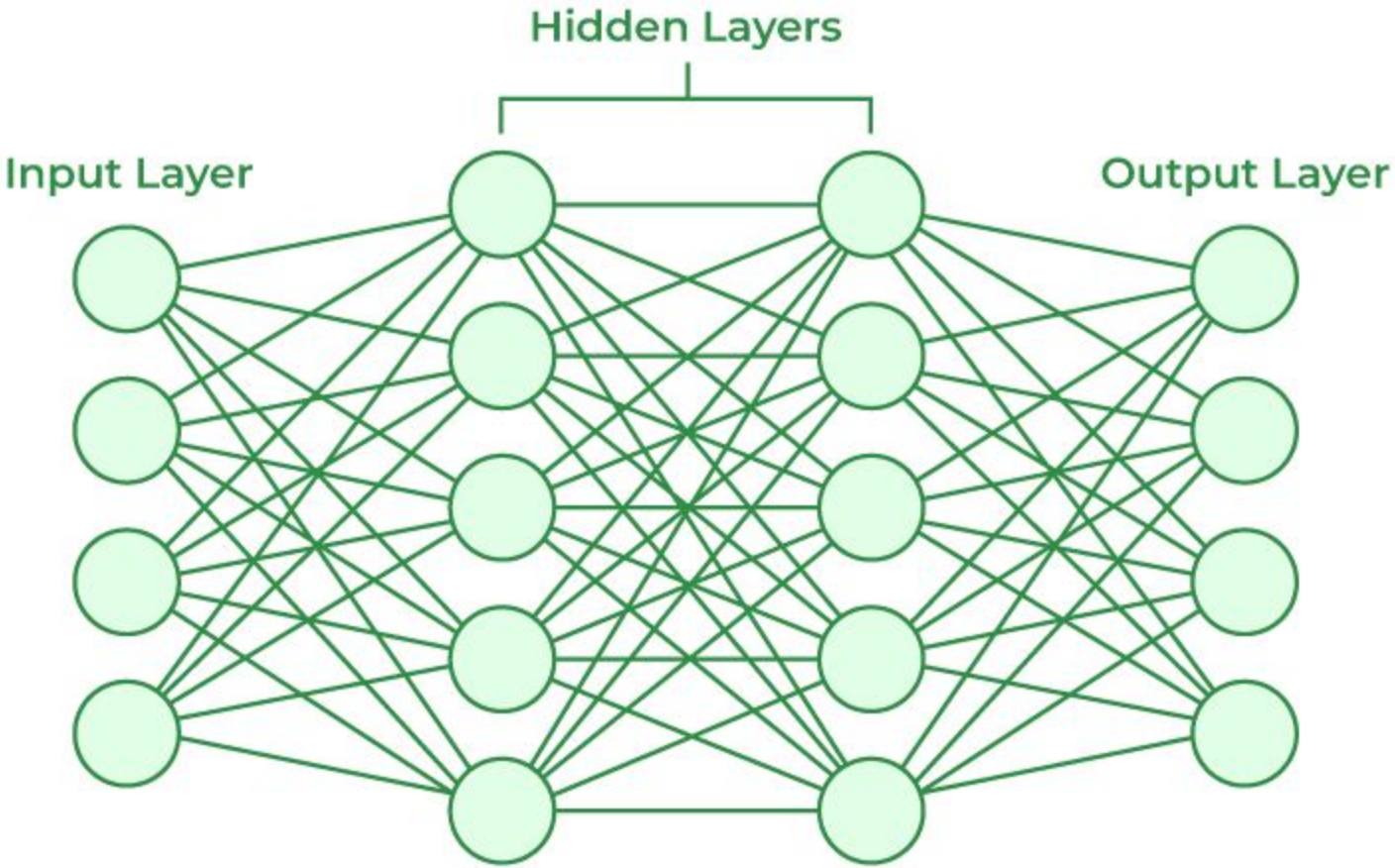
— Just like our brain learns from examples — a child learns to recognise a cat/dog after seeing many dogs and cats

— Like Machine Learning, Deep Learning models learn from data.

Neurons → artificial neurons

Synapses → weights

Learning → weight updates



The word *deep* comes from multiple hidden layers.

“Thank You”